



BIOLOGY CH.19 — ENZYMES & HORMONES — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway



Topics in this chapter:

1. Insulin & Diabetes
2. Digestive Enzymes
3. Adrenaline & Stress Response
4. Plant Hormones
5. Human Endocrine Glands
6. Enzyme Functions & Sources
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1. Insulin & Diabetes

- Insulin deficiency → Diabetes mellitus (high blood sugar). Insulin pancreas द्वारा produce होता है, chemical messenger जो cells को glucose absorb करने देता है. [Q1098]
- Insulin function: body में glucose और fat का use/storage regulate करता है. Glucose को bloodstream से cells तक ले जाता है. Insulin deficiency → hyperglycemia, Type 1 diabetes. [Q1107]
- Glucagon (alpha cells in islets of Langerhans से) → blood glucose को कम होने से बचाता है, sugar level बढ़ाता है. [Q1115]
- Insulin vs Glucagon: Insulin = sugar कम करता है, Glucagon = sugar बढ़ाता है. [Q1115]

2. Digestive Enzymes

- Amylase (salivary glands + pancreas): starch → simple sugars. Saliva में Alpha-Amylase (Ptyalin) होता है. [Q1099, Q1109, Q1125, Q1131]
- Pepsin (gastric glands): protein digest करता है. HCl pepsin को activate करता है. Pepsin protein → peptides में तोड़ता है. [Q1105, Q1106, Q1122]
- Trypsin (pancreas → small intestine): protein → amino acids. Inactive trypsinogen form में आता है. [Q1123, Q1138]
- Lipase (pancreas): fats/triglycerides → fatty acids + glycerol. Rice bran oil stabilization में भी use होता है. [Q1101]
- Lactase: lactose break करता है. Sucrase: sucrose → fructose + glucose. Maltase: maltose → glucose. [Q1138]
- Invertase: sucrose → fructose + glucose. Zymase: fermentation में sugars break करता है. [Q1106]

3. Adrenaline & Stress Response

- Adrenaline/Adrenalin/Epinephrine = "fight or flight" hormone. Adrenal medulla से secreted. Anger, embarrassment, worry के time body को stress से adjust करने में मदद करता है. [Q1100, Q1104]
- Adrenaline से heartbeat और breathing rate simultaneously increase होती है. [Q1104]

- Estrogen = female reproductive organs. Thyroxine = digestion, heart/muscle function, brain development. Testosterone = primary male hormone. [Q1100]

4. Plant Hormones

- Prolactin = pituitary gland hormone in mammals (milk production), plant hormone nahi hai. [Q1110]
- Plant hormones: Ethylene (fruit ripening, leaf dropping), Absciscic acid (stomata closure, growth inhibition, wilting, stress), Gibberellins (stem/leaf growth), Cytokinins (cell division), Auxins (cell elongation, phototropism). [Q1110, Q1111, Q1135]
- Absciscic acid (ABA) = "stress hormone". Growth inhibit karta hai, wilting/pores closing ke liye responsible, seed dormancy. [Q1111, Q1128, Q1130]
- Cytokinins = cell division promote karte hain, root apices/developing shoot buds/young fruits mein synthesize hote hain. Apical dominance overcome karte hain. [Q1127, Q1129]
- Auxin = cell elongation, phototropism (stem light ki taraf bend hota hai), prevents premature fall of leaves/flowers/fruits. Universal natural auxin = IAA (Indole-3-Acetic Acid). NAA = vegetative propagation mein, IBA = root induction. [Q1119, Q1120, Q1137]
- Gibberellins/Gibberellic acid = stem elongation, seed germination, fruit development, dwarfness control karte hain. Cold temperature mein zyada produce hote hain. [Q1132, Q1134]
- Ethylene = fruit ripening, leaf dropping. Natural hormone. [Q1110]
- Growth hormones kabhi promoter, kabhi inhibitor ki tarah kaam kar sakte hain - context dependent. [Q1133]

5. Human Endocrine Glands

- Pituitary gland (Hypophysis) = "Master gland". Brain ke base par pea-sized. Anterior lobe: ACTH, Prolactin, TSH, Growth Hormone. Posterior lobe: Oxytocin, Vasopressin. [Q1103]
- Human Growth Hormone (HGH) = anterior lobe of pituitary se secreted hota hai. [Q1103]
- Thymus gland = sirf puberty tak active rehti hai. Thymosins secrete karti hai jo lymphocyte production regulate karta hai. [Q1113]
- Pineal gland = "biological clock", melatonin secrete karti hai (sleep regulator). [Q1113]
- Hypothalamus = releasing/inhibiting hormones. [Q1113]
- Testes = Testosterone (androgens) secrete karte hain. Leydig cells dwara. Puberty mein boys ke appearance mein changes laata hai. [Q1121, Q1124]
- Ovaries = Estrogen produce karti hain. Menstrual cycle regulate karta hai. [Q1112]
- Thyroid gland = Thyroxine (T4) aur Triiodothyronine (T3). [Q1117]
- Thyroxine commonly T4 ke naam se jana jata hai. TSH T4 aur T3 production regulate karta hai. PTH serum calcium regulate karta hai. [Q1117]
- Thyroxine synthesis ke liye Iodine essential hai. Deficiency → Goitre. Iron = hemoglobin. Sodium = BP. Calcium = bone health. [Q1118, Q1126]

6. Enzyme Functions & Sources

- Enzymes = biological catalysts. Chemical reactions ki speed regulate karte hain without being consumed. [Q1108]
- Salivary glands se Amylase secreted hota hai. 3 pairs: Parotid (watery, amylase rich), Submaxillary, Sublingual. Total saliva: 1000-1500 ml/day. [Q1125]
- Gastric glands → Pepsin. Pancreas → Trypsin, Lipase, Amylase. Small intestine → Lactase, Sucrase, Maltase. [Q1105, Q1122, Q1123, Q1138]
- HCl (gastric glands) → pepsin ki action facilitate karta hai, acidic medium deta hai. [Q1122]

7. Hormone Deficiencies & Effects

- Ghrelin = "hunger hormone". 28-amino-acid peptide stomach se. Appetite stimulate karta hai, food intake badhata hai, fat storage promote karta hai. [Q1102]
- Leptin = food intake reduce karta hai. Resistin = obesity ko diabetes se link karta hai. [Q1102]
- Melanin = hair/skin color ka determinant. Melanocytes dwara produced. Keratin = hair/skin/nails ka protein. Carotene = fruits/vegetables mein orange/yellow/red pigment. [Q1114]
- Vasopressin (ADH) = osmotic balance, BP, sodium homeostasis control karta hai. [Q1115]
- Oxytocin = uterine contractions childbirth mein stimulate karta hai. [Q1098]

8. ⚠ Traps / Key Differences

- Prolactin = animal hormone, plant hormone nahi. [Q1110]
- Auxins = plant hormone, animal hormone nahi. [Q1136]
- Insulin = glucose regulation, glucagon = glucose raise. [Q1115]
- Amylase = starch digestion (saliva + pancreas). Pepsin = protein (stomach). Trypsin = protein (small intestine, pancreas). Lipase = fats. [Q1099, Q1105, Q1123]
- Thyroxine = T₄, TSH = thyroid stimulating, T₃ = triiodothyronine. [Q1117]
- Iodine = thyroxine ke liye essential. [Q1118, Q1126]
- Growth hormone = anterior pituitary. Oxytocin/Vasopressin = posterior pituitary. [Q1103]
- Hormones bacteria mein absent hote hain. [Q1116]
- IAA = universal natural auxin. NAA aur IBA synthetic/use-specific hain. [Q1137]